ModelIt!

Teacher's Guide

The Roller Coaster Equation

How Physics Turns Height Into Speed and Screams Into Science

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-PS3-1, MS-PS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS3-1, MS-PS3-2

Construct and interpret graphical displays of data to describe the relationships of kinetic energy to the mass of an object and to the speed of an object. Develop a model to describe that when the arrangement of objects interacting at a distance changes, different amounts of potential energy are stored in the system.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe energy transformations between potential and kinetic energy.
Disciplinary Core Idea	**PS3.A Definitions of Energy / PS3.B Conservation of Energy** — Kinetic energy is proportional to mass and speed squared. Energy is conserved in transformations.
Crosscutting Concept	**Energy and Matter** — Students trace energy as it flows and transforms between potential and kinetic forms throughout the system.

CAST Testing Alignment

MULTIPLE CHOICE:

A roller coaster car is at the top of a 30-meter hill. As it rolls to the bottom, what happens to its potential and kinetic energy?

CONSTRUCTED RESPONSE:

Using the principle of energy conservation, explain why a roller coaster's second hill can never be taller than its first hill (ignoring external energy sources).

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Roller coasters are elegant demonstrations of energy conservation. Every scream, every thrill, and every stomach drop is the result of energy transforming between two forms: potential and kinetic.

Potential Energy

Gravitational potential energy depends on height and mass: PE = mgh (mass × gravity × height). At the top of the first hill, the coaster has maximum PE. The chain lift does work against gravity to give the coaster this stored energy. A 500 kg coaster at 50 meters has PE = 500 × 9.8 × 50 = 245,000 Joules.

Kinetic Energy

Kinetic energy depends on mass and speed: KE = ½mv^2 (half × mass × speed squared). Notice that speed is SQUARED — doubling speed quadruples kinetic energy. This is why the first drop feels so intense: small changes in height create large changes in speed.

Conservation of Energy

In an ideal system (no friction), total energy = PE + KE = constant. At the top: high PE, low KE. At the bottom: low PE, high KE. The energy doesn't disappear — it transforms. In real coasters, some energy is lost to friction and air resistance as heat.

Real Coaster Design

Engineers use these equations to design every hill, loop, and turn. The first hill must be the tallest because the coaster can never regain that height without additional energy input. Loop-the-loops require a minimum speed at the top to maintain contact with the track.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify coaster height, coaster mass, potential energy, and kinetic energy as system components.

E — Establish Relationships
Students determine that height positively affects PE, and as PE decreases (going downhill), KE increases — they trade back and forth.

V — Visualize & Model
Students build the energy transformation model showing the PE-KE seesaw.

E — Evaluate Outcomes
Students compare different heights and masses to see how they affect the energy balance.

R — Revise & Extend
Students add friction to see why real coasters lose energy and eventually stop.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"No engine. No fuel. Just height. How does a roller coaster hit 70 mph?"*

Watch For

Show a first-person POV video of a big coaster drop.

Slide 2: Learning Objectives

3 min

Say: *"Today we crack the code of how height becomes speed."*

Watch For

Define PE and KE with quick physical demos.

Slide 3: Big Question

3 min

Say: *"The chain lift takes you up... then lets go. Everything after is pure physics."*

Watch For

Poll: Where on the coaster do you have the most energy?

Slide 4: LEVER Framework

2 min

Say: *"We'll model the energy seesaw that makes coasters work."*

Watch For

Preview the modeling process.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key energy players in a roller coaster?"*

Watch For

Sort components. Discuss what's controlled vs. what changes.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the key: PE and KE are on opposite ends of a seesaw."*

Watch For

Demonstrate with a ball on a ramp. When one goes up, the other goes down.

Slide 7: Activity 3: Simulation

10 min

Say: *"Watch PE and KE trade places as the coaster moves."*

Watch For

Students run scenarios and sketch the PE-KE graphs.

Slide 8: Discoveries

5 min

Say: *"The total energy stays CONSTANT. Height and speed are the same energy in different forms."*

Watch For

Discuss: Why can't the second hill be taller?

Slide 9: STEM Challenge

12 min

Say: *"Design the ultimate coaster — but it must obey the laws of physics!"*

Watch For

Teams design and test with marble tracks or simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Coaster Height, Coaster Mass
- **INTERNAL:** Potential Energy, Kinetic Energy (Speed)

Relationships

- **Coaster Height to Potential Energy:** POSITIVE (+) — Higher position means more stored gravitational energy.
- **Coaster Mass to Potential Energy:** POSITIVE (+) — More mass at the same height means more stored energy.
- **Potential Energy to Kinetic Energy:** POSITIVE (+) (inverse seesaw) — As PE decreases going downhill, KE increases by the same amount — energy is conserved.

Key Discoveries

- At the top of the hill: maximum potential energy, minimum kinetic energy
- At the bottom: maximum kinetic energy, minimum potential energy — the energy TRANSFORMED
- The total energy stays the same (conservation!) — height energy becomes speed energy
- More mass means more of BOTH types of energy, but the speed at the bottom is the same

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Heavier coasters go faster"	Mass cancels out in energy calculations! A heavy coaster and a light coaster dropped from the same height reach the same speed (ignoring friction). They have different amounts of energy, but the speed is identical.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The coaster has no energy at the top"	At the top, the coaster is moving slowly (low KE) but has MAXIMUM potential energy stored in its height. It's like a fully charged battery — the energy is there, just stored.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Energy is used up on the ride"	Energy is never created or destroyed — only transformed. 'Used up' energy actually became heat through friction. The total energy of the universe stays the same.